Dibyendu Das

PhD Student in Mathematics

Indian Statistical Institute, Bangalore • Supervisor: Prof. Maneesh Thakur

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Algebraic Groups | Automorphism Groups | Associative Algebras

Research Interests

Algebraic Groups and related structures. My primary focus is on the algebraic and geometric properties of **automorphism groups of finite-dimensional associative algebras**, including their R-equivalence classes, solvability, and connections to isolated hypersurface singularities.

Publications & Preprints

[1] On R-Equivalence of Automorphism Groups of Associative Algebras *(Submitted)*

[2] On Solvability of Automorphism Groups of Commutative Algebras *(Accepted in **Journal of algebra**)*

Education

PhD in Mathematics — **2022 – Present**
Indian Statistical Institute, Bangalore
Supervisor: Prof. Maneesh Thakur · Thesis area: Automorphism groups of algebras

M.Math (Master of Mathematics) — **2019 – 2021**
Indian Statistical Institute, Bangalore
88.25%

B.Sc in Mathematics — **2016 – 2019**
St. Xavier's College, Kolkata
84.39%

Higher Secondary (Class XII) — **2014 – 2016**
Jangalpara BCKM High School (WBCHSE)
93.8%

Secondary (Class X) — **2014**
KCS High School (WBBSE)
89.5%

National Examinations & Entrance Results

Examination	Year	Rank / Result
IIT JAM Mathematics	2019	AIR **145**
CSIR-NET Lectureship	2019	AIR **47**
ISI M.Math Entrance	2019	AIR **6**
GATE Mathematics	2021	AIR **339**
CSIR-NET Mathematical Science (June)	2021	AIR **85**
GATE Mathematics	2022	AIR **13**
ISI JRF Entrance	2022	AIR **8**
NBHM PhD Entrance	2022	**Top 40**

Also selected for: INSPIRE Fellowship (Stage 2) | JRF UGC NET 2021 | NBHM Doctoral Fellowship 2022

Awards & Scholarships

INSPIRE Scholarship — *DST, Govt. of India (Top 1% in Class XII)* — **2016 – 2020**

Ram Ghosh Encouragement Award — *St. Xavier's College, Kolkata (70%+ in Semesters I–IV)* **2016 – 2018**

Dhruva Jyoti Bhattacharya Memorial Gold Medal — *St. Xavier's College (Highest marks in Mathematics, B.Sc)* **2016 – 2019**

ISI Scholarship — *Indian Statistical Institute* **2020 – 2021**

Teaching Experience

Teaching Assistant — Algebraic Topology — *ISI Bangalore, B.Math 3rd Year (Instructor: Charanya Ravi)* **Fall 2023**

Teaching Assistant — Probability III — *ISI Bangalore, B.Math 3rd Year (Instructor: Vivek Kumar)* **Fall 2024**

Teaching Assistant — Topology I — *ISI Bangalore, B.Math 2nd Year (Instructor: Maneesh Thakur)* **Spring 2025**

Teaching Assistant — Probability II — *ISI Bangalore, BSDS 2nd Year (Instructor: Antar Bandyopadhyay)* **Fall 2025**

Teaching Assistant — Geometric Group Theory — *ISI Bangalore, B.Math 3rd Year (Instructor: Maneesh Thakur)* **Spring 2026**

Teaching Assistant -Algebraic Number Theory — *ISI Bangalore, B.Math 3rd Year, M.Math 2nd Year (Instructor-Maneesh Thakur)* **Fall 2026**

Evaluator — Regional Mathematical Olympiad (Karnataka) — *Homi Bhabha Centre for Science Education* **2023**

Lecture Series (Advanced Graduate)

Affine Algebraic Groups and their Structures — *ISI Bangalore* **Jan – Jun 2023**

Trialitarian Algebras and Book of Involution — *ISI Bangalore* **Jan – Aug 2024**

Schemes — *ISI Bangalore* **Aug – Nov 2025**

Theory of higher degree forms and their automorphism groups — *ISI Bangalore* **Jan 2026 – May 2026**

Research Talks

Algebraic Groups and Automorphism Groups of Algebras — *PDF-RS Symposium, ISI Bangalore* **Oct 2023**

On Automorphism Groups of Algebras — *JRF to SRF Evaluation, ISI Bangalore* **May 2024**

R-Equivalence on Automorphism Groups of Associative Algebras — *RDGT 2025, IIT Bhubaneswar* **Jul 2025**

Solvability of Automorphism Groups and Isolated Hypersurface Singularities — *PDF-RS Symposium, ISI Bangalore* **Mar 2026**

On R-Equivalence of Automorphism groups of Associative Algebras — Colloquium Talk — *ISI Bangalore* **Mar 2026**

Solvability of Automorphism groups of Commutative Algebras — *PGTC 2026, University of Manchester, UK.* **June 2026**

Solvability of Automorphism groups of Commutative Algebras and Isolated Hypersurface Singularities — *GGI Seminar 2026, University of Essex, UK.* **July 2026**

On R-equivalence of Automorphism groups of Associative algebras — *AGGITatE 2026, INI Satelite Program, University of Essex, Uk* **July 2026**

Solvability of Automorphism groups of commutative algebras and Isolated hypersurface singularities — *Workshop in Group Theory 2026, IISER Pune* **October 2026**

Conferences & Schools Attended

- **Annual Seminar "ANALYTICA",** St. Xavier's College, Kolkata *2017, 2018, 2019*
- **ATM School on Advanced Commutative Algebra**, IIT Gandhinagar (NCM) *Jan 2021*
- **Zoom Algebraic Geometry Seminar,** University of Essex, UK *Jun 2021*
- **AIS on Birational Geometry**, Chennai Mathematical Institute (NCM) *May–Jun 2023*
- **Zariski Dense Subgroups, Number Theory and Applications**, ICTS Bangalore *Jan 2024*
- **AIS on Representation Theory of Quivers and BGG Category $\mathcal{O}$**, IIT Kanpur (NCM Online) *Dec 2024*
- **Recent Developments in Group Theory (RDGT 2025)**, IIT Bhubaneswar *Jul 2025*
- **Post Graduate Group Theory Conference (PGTC 2026)**, University of Manchester *June-July 2026*
- **Algebraic groups, geometry, invariants and related topics (GGI 2026/AGGITatE 2026)**, University of Essex INI Satellite Program *July 2026*
- **Workshop in Group Theory 2026 (WGT 2026)**, IISER Pune *October 2026*

Undergraduate Project

A Study on Solvable Groups — *B.Sc. Dissertation, St. Xavier's College (Supervisor: Prof. Rabiul Islam)* **Dec 2018 – Jun 2019**
Studied solvable and nilpotent groups and applied representation theory to establish Burnside's theorem.

References

Prof. Maneesh Thakur
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Prof. B Sury
NASI Senior Scientist, TSMU
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